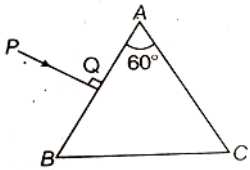
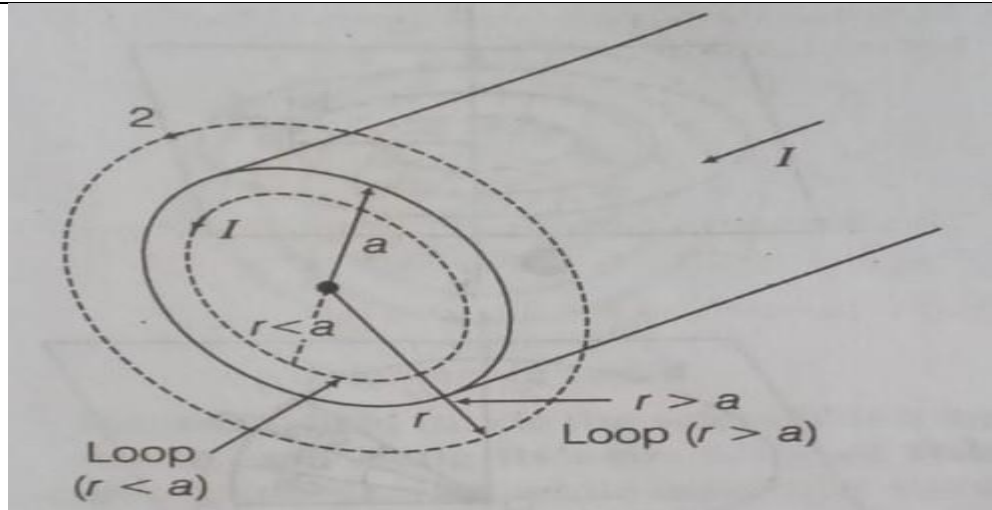


	MARKING SCHEME
1	a
2	d
3	b
4	d
5	c
6	a
7	b
8	d
9	c
10	b
11	C
12	C
13	b
14	a
15	b
16	a
17	Correct definition.....1/2 Mobility $\mu = \frac{v_d}{E} = \frac{e\tau}{m}$1 Unit $m^2 V^{-1} s^{-1}$1/2
18	$R = \frac{V}{I_g} - G = \frac{15}{5 \times 10^{-3}} - 50 = 2950 \Omega$1½ 2950 Ω is to be connected in series with galvanometer to convert it into a desired voltmeter.....1/2
19	$\frac{I_{min}}{I_{max}} = \frac{9}{25}$ $\frac{(a_1 - a_2)^2}{(a_1 + a_2)^2} = \frac{9}{25}$1/2 $\frac{a_1}{a_2} = \frac{4}{1}$1/2 $\frac{I_1}{I_2} = \frac{16}{1}$1/2 Ratio of widths of the slit $\frac{d_1}{d_2} = \frac{I_1}{I_2} = \frac{16}{1}$1/2 OR $\frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2}$1/2 $\lambda_2 = \frac{\beta_2}{\beta_1} \times \lambda_1 = \frac{8.1 \times 10^{-3}}{7.2 \times 10^{-3}} \times 630 \times 10^{-9} \text{ m}$1/2 $\lambda_2 = 708.75 \times 10^{-9} \text{ m} = 708.75 \text{ nm}$1
20	Correct definition.....1 Correct graph.....1
21	Correct graph.....1 Correct explanation1
22	Correct diagram1 Two points2

23	Correct circuit diagram.....1 Explanation of working.....1 Input and output waveforms.....1
24	 $A=60^0$ $i=0^0$.....1/2 At the interface AB $\frac{\sin i}{\sin r} = \frac{\mu_g}{\mu_a}$ $r=0^0$.....1/2 At the interface AC $\frac{\sin i}{\sin e} = \frac{\mu_g}{\mu_a}$ $\sin e = \frac{2}{\sqrt{3}} \times \sin 60^0$.....1/2 $e=90^0$1/2 angle of deviation=30^0.....1
25	Correct statement.....1 $\lambda = \frac{q}{l}$.....1/2 $\Phi = \frac{q}{\epsilon_0}$1/2 $E \times 2\pi r l = \frac{\lambda l}{\epsilon_0}$.....1/2 $E = \frac{\lambda}{2\pi r \epsilon_0}$1/2 OR Correct definition.....1/2 Unit Nm^2C^{-1}.....1/2 $\Phi = \frac{q}{\epsilon_0}$ $2 \times E \int dS = \frac{q}{\epsilon_0}$1/2 $E \times 2A = \frac{q}{\epsilon_0}$1/2 $E = \frac{q}{2A\epsilon_0}$ $E = \frac{\sigma A}{2A\epsilon_0}$.....1/2 $E = \frac{\sigma}{2\epsilon_0}$1/2
26	Diagram.....1/2 Explanation of formation of depletion region and barrier potential1$\frac{1}{2}$ Any two differences.....1

27

For $r < a$ Current passes through the cross-section of radius r is $I_1 = \frac{I}{\pi a^2} \times \pi r^2$

.....1/2

 $B \oint dl = \mu_0 I_1$ 1/2

 $B \times 2\pi r = \frac{\mu_0 I}{a^2} \times r^2$ 1/2

 $B = \frac{\mu_0 I}{2\pi a^2} \times r$ 1/2
For $r > a$
 $B \oint dl = \mu_0 I$ 1/2

 $B \times 2\pi r = \mu_0 I$
 $B = \frac{\mu_0 I}{2\pi r}$ 1/2

28

For dielectric

 $C = \frac{A\epsilon_0}{(d-t+\frac{t}{k})}$ 1

For metallic slab

 $C = \frac{A\epsilon_0}{(d-t)}$ 1

In second case capacitance will be more.....1/2

Correct reason.....1/2

29

- i)d
- ii)d
- iii)c
- iv)d
- OR
- iv)b

30

- i)d

	ii)c iii)a iv)b OR iv)a
31	i) Electric field produced inside the wire is $E = V/l$ Force on an electron $a = -eE$1/2 Acceleration of each electron $= -eE/m$1/2 Drift velocity $v_d = -eE\tau/m$1 Current in the conductor $I = neAv_d$1/2 $I = \frac{nA\tau Ve^2}{ml}$ $I = VR$1/2 ii) Electric field remains constant which is equal to V/l . current density and hence drift speed change2 OR i) correct statements2 ii) $I_3 = I_1 + I_2$1/2 for loop 1 $4I_1 - 3I_2 = -1$1/2 For loop 2 $3I_2 + 2I_3 = 3$1/2 $I_1 = 2/13A$1/2 $I_2 = 7/13A$1/2 $I_3 = 9/13A$1/2
32	i) Correct diagram1 principle of working of ac generator2 ii) correct expression1 iii) mechanical energy of rotation of armature coil1 OR i) correct diagram1 principle of transformer1/2 correct expression $\frac{V_1}{V_2} = \frac{N_1}{N_2}$1/2 For ideal transformer, input power = output power1/2 $\frac{V_1}{V_2} = \frac{I_2}{I_1}$1/2 ii) any two sources of energy losses1 iii) correct explanation1
33	(i) Correct diagram1 $m = \frac{\beta}{\alpha} = \frac{\tan\beta}{\tan\alpha} = m_0 \times m_e$1 $m = \frac{v_0}{u_0} \left(1 + \frac{D}{f_e}\right)$1 (ii) $\frac{1}{f_0} = \frac{1}{v_0} - \frac{1}{u_0}$, $v_0 = 7.5cm$1

	$m = -30$.....1 OR (i) Correct diagram and derivation.....3 marks (ii) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ for lens L_1 $V = 30\text{cm}$.....1/2 marks for lens L_3, object distance infinity1/2 marks separation between L_1 and L_2, $30 + 10 = 40\text{cm}$.....1/2 marks the distance between L_2 and L_3 may take any value.....1/2 marks
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